

Efficiency gain and bidirectional operation of quantum engines with decoupled internal levels

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We present a mechanism for efficiency increase in quantum heat engines containing internal energy levels that do not couple to the external work sink. The gain is achieved by using these levels to channel heat in a direction opposite to the one dictated by the second law. No quantum coherence, quantum correlations or ergotropy are required. A similar mechanism allows the engine to run “in reverse” and still produce useful work. We illustrate these ideas using a simple quantum Otto cycle in a coupled-spin system. We find this engine also exhibits other counterintuitive phenomenology. For example, its efficiency may increase as the temperature difference between the heat baths decreases. Conversely, it may cease to operate if the hotter bath becomes too hot or the colder bath too cold.

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I. INTRODUCTION

There has been a resurgence of interest in quantum thermodynamics [1], in particular in microscopic engines operating in the quantum regime [2,3]. A major goal is to understand how these quantum engines may differ from their classical counterparts. It is already known, for example, that energy-basis coherence may boost engine power [4,5], a phenomenon that has recently been observed experimentally [6]. A closely related mechanism [8,9] is to exploit ergotropy—energy extractable by cyclic unitary rotations, without changing a system’s entropy [7] to boost engine efficiency. This may also occur when at least one heat reservoir interacting with the engine is not in thermal equilibrium [8–11], in particular when it is, actually a quantum measuring device [12–14]. In contrast, no firm link has been found between efficiency gains and the presence of quantum correlations between subsystems of an engine’s working medium [15–25].

In this article, we point out a simple alternative mechanism by which a quantum engine operating between two thermal heat baths may also achieve increased efficiency. It applies in scenarios when the working medium has discrete internal levels that do not couple to the external work sink. These levels may be made to channel heat in a reversed direction (i.e., from the cold bath to the hot), thus allowing more heat to be converted to work via the coupled levels. A similar mechanism allows the engine to produce work while running the same thermodynamic cycle in reverse (Fig. 1). We also identify associated effects regarding such engines’ responses to changes in the baths’ temperatures. For example, they may sometimes become less efficient, or even stop working altogether, when the temperature difference increases.

II. OTTO CYCLES

We restrict our discussion to the context of quantum Otto cycles [26–30], although most of the concepts are more gener-

ally applicable. These cycles use a working medium described by a Hamiltonian $H(\lambda)$, with energies $E_n(\lambda)$, depending on some external parameter λ . They consist of two “isochoric” [31] strokes and two (quantum) adiabatic strokes. In the former, λ is fixed at values λ_a or $\lambda_b < \lambda_a$ while the system equilibrates with an external thermal reservoir at respective temperatures T_a, T_b . The corresponding thermal states have eigenvalues $p_n^j \equiv e^{-E_n^j/T_j}/Z_j$, where $E_n^j \equiv E_n(\lambda_j)$ and $Z_j = \sum_n e^{-E_n^j/T_j}$. In the adiabatic strokes, the system is isolated from the reservoirs while λ is varied back or forth between λ_b and λ_a . Supposing that no level crossings occur, and that this variation is sufficiently slow, the populations of each level remain constant by the quantum adiabatic theorem. For simplicity, in this article we mostly consider this idealized—but experimentally feasible [32–34]—regime. In some situations, however, the same evolution can be achieved for λ driven in finite time, see Sec. III.

In these cases it is straightforward to calculate the average heat and work exchanges [2,3]. During the isochoric strokes H is fixed, so only heat is exchanged:

$$Q_a = \sum_n E_n^a \Delta p_n; \quad Q_b = \sum_n -E_n^b \Delta p_n \quad (1)$$

where $\Delta p_n \equiv p_n^a - p_n^b$. During adiabatic strokes entropy is fixed, and only work is exchanged. The overall extracted work is

$$W_{\text{cycle}} = \sum_n \Delta p_n (E_n^a - E_n^b). \quad (2)$$

We will find it useful below to interpret each term in these sums as a separate energy flow, e.g., to view $q_n^a \equiv E_n^a \Delta p_n$ as the heat exchanged with bath a via level n .

It is sometimes claimed [30,35] that the efficiency $\eta = W_{\text{cycle}}/Q_{\text{hot}}$ of these idealized Otto cycles must equal $\eta_0 = 1 - \frac{1}{r}$, where $r = \lambda_a/\lambda_b > 1$. However, in general this only holds for working media whose energy gaps *all shift in proportion to λ* , i.e., whose energies satisfy

$$E_n(\lambda) = A(\lambda) + c_n \lambda, \quad (3)$$

for constants c_n and some function $A(\lambda)$ independent of n . Examples include a harmonic oscillator with variable

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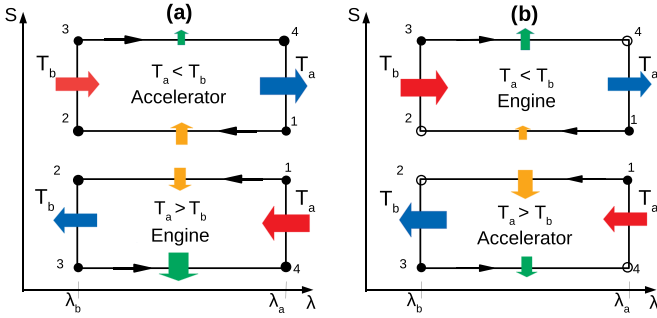


FIG. 1. (a) Otto cycles for a quantum system whose energy levels all follow Eq. (3) during the adiabatic (horizontal) strokes. In this case, a heat engine must operate counterclockwise. A clockwise cycle can represent either an accelerator [14] (depicted) or a refrigerator (not depicted). (b) If there are also “idle” levels, independent of λ , then an engine may also operate clockwise, and an accelerator counterclockwise. Filled (respectively, empty) circles represent thermal equilibrium (respectively, nonequilibrium) states. Other operation modes are possible in either cycle sense, see Fig. 3.

frequency [36], uncoupled spins in a variable magnetic field [26,27] and even some coupled spin systems [37]. For such systems, an adiabatic stroke maps an initial thermal state to another at a different temperature. Thus, in these cases, the quantum notion of an adiabatic evolution coincides with the thermodynamic one [28].

In general, however, a quantum adiabatic stroke will drive a medium out of equilibrium [28], although it will still remain in a passive [7], energy-diagonal state. As we now show, this can affect η . Consider for instance a minimal departure from the previous scenario, where now only a subset $\{E_n\}_{n \in \mathcal{W}}$ of the energy levels are “working,” i.e., vary with λ as in Eq. (3), while the remainder are “idle,” or independent of λ . For simplicity, we assume $A(\lambda) \equiv 0$ [38]. Assuming for the moment $T_a > T_b$, so $Q_a = Q_{\text{hot}} > 0$, then Eq. (2) implies:

$$\frac{\eta}{\eta_0} = 1 - \frac{\sum_{n \notin \mathcal{W}} q_n^a}{Q_a}. \quad (4)$$

Equation (4) is our first main result. It says that, for $T_a > T_b$, enhanced efficiency $\eta > \eta_0$ occurs if $\sum_{n \notin \mathcal{W}} q_n^a < 0$, i.e., if on the whole idle levels channel heat *into* the hot reservoir [compare Figs. 2(a) and 2(b)]. This makes physical sense: For idle levels, $q_n^a = -q_n^b$, i.e., all the heat they absorb from one bath is deposited into the other. Thus, any heat flux from a to b via these levels is wasted for the purpose of generating work. Conversely, for a given fixed Q_a , reversing this flux allows more heat to flow via working levels, increasing η . For idle levels with $E_n > 0$, such a reversal is achieved when their equilibrium population at a (i.e., at higher T and λ) is lower than at b and vice versa for those with $E_n < 0$. Note this need not violate the second law, which only constrains the *overall* heat transfers in a cycle.

If $T_b > T_a$, then the roles of hot and cold bath are exchanged, i.e., $Q_b = Q_{\text{hot}} > 0$. In this case the cycle runs clockwise (Fig. 1) and ordinarily we expect it to represent either a refrigerator (where heat flows out of the colder bath and into the hotter) or an *accelerator* [14] (where heat flows out of the hotter bath and, at a higher rate, into the colder).

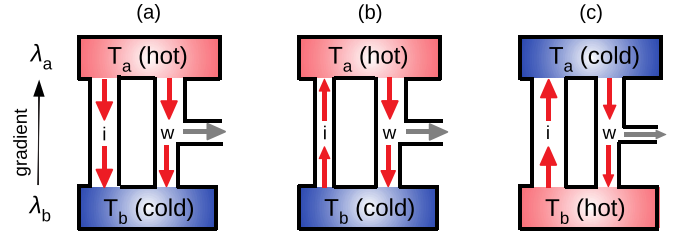


FIG. 2. Energy transfer diagrams for various heat engine regimes. The left and right channels represent respectively heat flowing via idle (i) and working (w) levels. (a) An “ordinary” engine, where both channels flow from the hot bath to the cold. (b) An engine cycling in the “ordinary” sense [Fig. 1(a)], but with enhanced efficiency $\eta > \eta_0$. (c) A “counter-rotating” engine, as in Fig. 1(b). In regimes [(b) and (c)], one channel must flow in the “wrong” sense, i.e., from cold to hot.

However, now

$$\frac{\eta}{\eta_0} = r \left(\frac{\sum_{n \notin \mathcal{W}} q_n^b}{Q_b} - 1 \right) = -r \frac{\sum_{n \in \mathcal{W}} q_n^b}{Q_b}. \quad (5)$$

Equation (5) is our second main result. It shows that, for $T_b > T_a$, a heat engine requires $\sum_{n \in \mathcal{W}} q_n^b < 0$, i.e., now the *working* levels must channel heat in the “wrong” sense [Fig. 2(c)]. We show below that such a “counter-rotating” engine is not only possible but occurs quite generally, for example, when the ground state is idle. These engines can also operate at efficiency $\eta > \eta_0$ if the reverse heat flux is sufficiently strong ($> Q_b/r$).

III. MODEL

In the remainder of this article we study these behaviors, along with other associated effects, using a simple model [20,39] of two spin-1/2 particles coupled via the Heisenberg interaction and placed in an external static magnetic field $\vec{h} = h\hat{z}$:

$$H = 2J \vec{\sigma}_1 \vec{\sigma}_2 + h(\sigma_1^z + \sigma_2^z) - 2J \quad (6)$$

with $\vec{\sigma}_i = \{\sigma_i^x, \sigma_i^y, \sigma_i^z\}$ the Pauli matrices for site i and $J \geq 0$ the antiferromagnetic exchange coupling [40]. This Hamiltonian is diagonal in the total spin basis. The constant $-2J$ is added so that its energies have the simple forms $-8J$, $-2h$, 0 , and $+2h$. For definiteness, we will refer to them as $E_{1,2,3,4}$, in this order, regardless of the relative magnitudes of J and h . We are interested in cycles where the field strength $h > 0$ plays the role of λ . Note E_{2-4} satisfy Eq. (3), with $A(h) \equiv 0$ and $c_{2,3,4} = -1, 0, +1$. However, E_1 is (nontrivially) idle if $J > 0$. Note, finally, that, since the energy eigenbasis is independent of h , a time-dependent variation of h does not change energy level populations. In other words, “adiabatic” strokes can here in fact be realized in finite time.

For this model:

$$\begin{aligned} Q_a &= -8J(p_1^a - p_1^b) + 2h_a(f_b - f_a) \\ Q_b &= -8J(p_1^b - p_1^a) + 2h_b(f_a - f_b) \\ W_{b \rightarrow a} &= 2(h_a - h_b)f_b > 0 \\ W_{a \rightarrow b} &= -2(h_a - h_b)f_a < 0 \\ W_{\text{cycle}} &= W_{b \rightarrow a} + W_{a \rightarrow b} = 2(h_a - h_b)(f_b - f_a), \end{aligned} \quad (7)$$

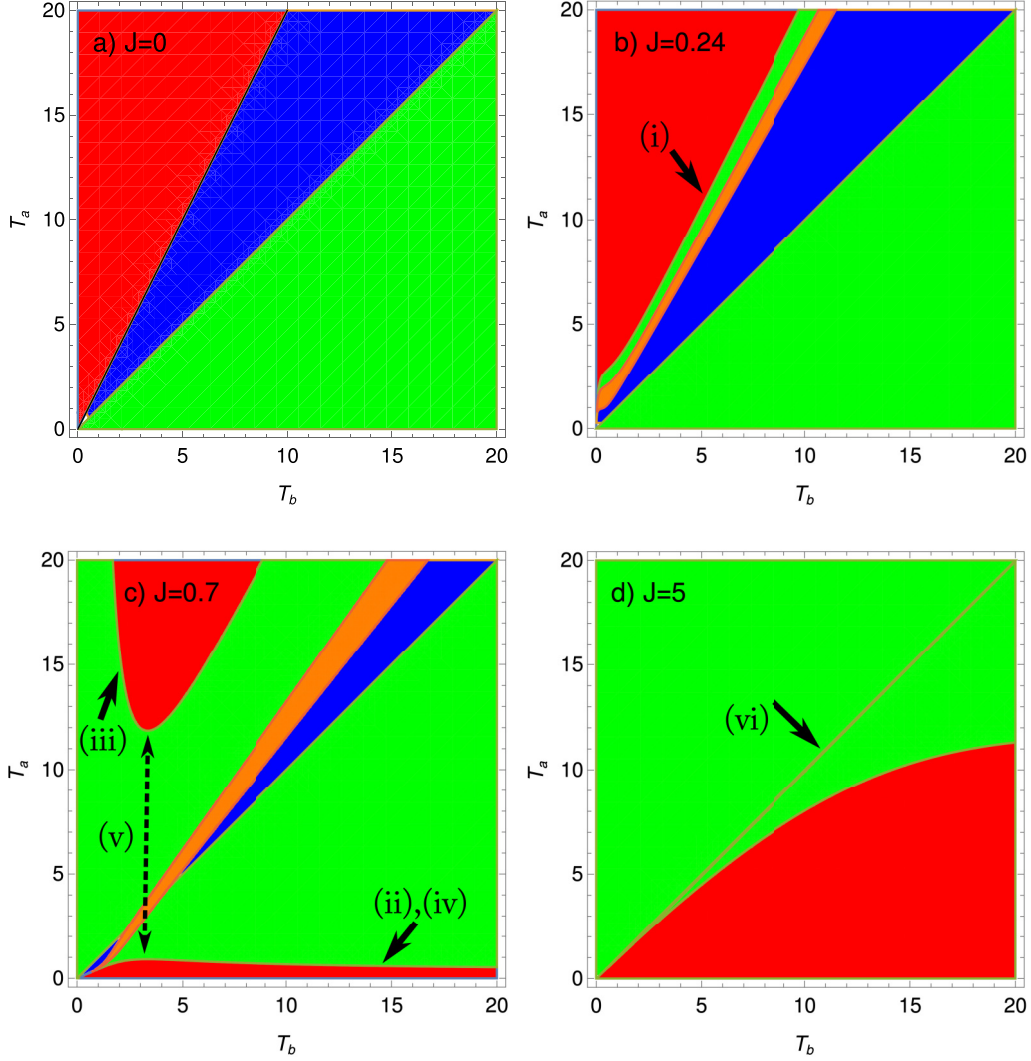


FIG. 3. Temperature ranges where Eq. (7) describes a refrigerator [blue (darkest gray)], engine [red (darker gray)], heater [orange (lighter gray)], or accelerator [green (lightest gray)] for various coupling strengths J . Field values are $h_a = 2$, $h_b = 1$. J , T , and h are in the same (arbitrary) energy units [40]. Several remarkable phenomena [(i)–(vi)] can be inferred (see text).

where

$$f_j(h, J, T) \equiv p_2^j - p_4^j = \frac{2 \sinh 2h_j/T_j}{1 + e^{8J/T_j} + 2 \cosh[2h_j/T_j]} \quad (8)$$

Note that $W_{\text{cycle}} > 0$ if and only if $f_b > f_a$.

Although the effects we are interested in only occur for coupled spins ($J > 0$), it is useful to review what happens when $J = 0$ [26–28]. In this case $f(h, J, T)$ reduces to $\sinh(2h/T)/[1 + \cosh(2h/T)]$, which increases monotonically with h/T , so $f_b > f_a \iff T_a > rT_b$. The same condition also implies $Q_a > 0$, $Q_b < 0$. Thus, the cycle operates as a heat engine only if it is performed counterclockwise on an entropy-field diagram [Fig. 1(a)] [41].

Its efficiency is $\eta_0 = 1 - \frac{1}{r}$, as expected. Since $T_a > rT_b$, clearly $\eta_0 < 1 - \frac{T_b}{T_a} \equiv \eta_{\text{Carnot}}$.

If we decrease T_a so that $f_a > f_b$, then Q_a , Q_b , W_{cycle} all change sign, and the cycle becomes clockwise. Two other regimes of operation are then possible: For $T_b < T_a < rT_b$ we have a refrigerator, and for $T_a \leq T_b$ an accelerator [Fig. 1(a)],

since bath b is now the hot bath. These behaviors mirror those of an ideal-gas Otto cycle.

IV. COUNTERINTUITIVE PHENOMENOLOGY

For coupled spins, the existence of the “idle” level $E_1 = -8J < 0$ allows the two unconventional engine regimes depicted in Figs. 2(b) and 2(c) to become possible. These effects are part of a wider pattern of phenomena that run counter to (at least these authors’) usual expectations for a thermal machine. To help appreciate this, in Fig. 3 we plot, for several values of J , the ranges of temperatures T_a , T_b for which the cycle operates in each of the four regimes allowed by Thermodynamics [14]: engine (E), refrigerator (R), accelerator (A), or heater (H) (where work is converted into heat entering both baths). As noted above, for $J = 0$ only the first three possibilities occur, separated by the lines $T_a = rT_b$ and $T_a = T_b$ [Fig. 3(a)]. As J increases, however, new features appear:

(i) For low J , the E and R zones shrink, while a second A zone appears, as well as the previously forbidden heater regime [Fig. 3(b)].

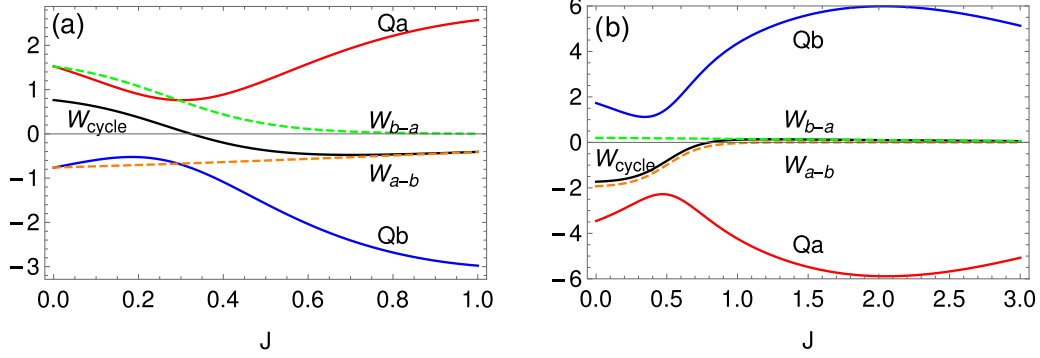


FIG. 4. Work and heat as a function of the coupling J for $h_a = 2$, $h_b = 1$. (a) $T_a = 5$, $T_b = 1$. As J grows, the cycle transitions from an engine to an accelerator. (b) $T_a = 1$, $T_b = 10$. As J grows, the cycle transitions from an accelerator to a (very inefficient) engine.

(ii) For sufficiently large J , a second E zone appears, within the region where $T_b > T_a$ [Figs. 3(c) and 3(d)]. This is the counter-rotating (clockwise) engine regime, described in our second main result, and in Figs. 1(b) and 2(c). Note that, for this specific model, this phenomenon was noticed, but not fully explored, in Ref. [39].

(iii) “Ordinary” (counterclockwise) engines are also still achievable at these J ’s, within the region where $T_a > rT_b$. However, for any finite T_a , an engine becomes *impossible* for sufficiently small T_b . This is remarkable; we usually expect that the lower the cold bath temperature the *easier* it is to run an engine.

(iv) Conversely, for $T_b > T_a$ an engine becomes impossible when T_b becomes too *high*. [Notice the downward slope of the lower E zone in Fig. 2(c), for large-enough T_b]. Thus, the hot bath can become too hot.

(v) The two E zones are separated by a *gap* in T_a , within which no heat engine is possible, regardless of T_b .

(vi) For larger J , the H and R zones collapse to the line $T_a = T_b$ [Fig. 3(d)].

V. PHYSICAL INTERPRETATION

This phenomenology can again be physically understood from the level structure, in particular from the fact that level $E_1 = -8J$ is “idle.” For example, the condition $J > \frac{h_a}{4}$, valid in Figs. 3(c) and 3(d), implies that E_1 remains the ground state throughout the cycle. In particular, it concentrates most of the system’s population after thermalization with bath a . Since, however, it does not shift with h , then for sufficiently low $T_a \ll T_b$ the invested work $|W_{a \rightarrow b}|$ must become smaller than the extracted work $W_{b \rightarrow a}$. This explains the “counter-rotating” engine [property (ii)]. Note also that, by Eqs. (7) and (8), both work exchanges fall rapidly with J , so we might expect engines in this regime to have a very small output. While this does happen in some cases [e.g., Fig. 4(b)], a nonvanishing work output is nevertheless still possible in the limit where T_a is very small, see the inset in Fig. 5(b).

Property (iii) follows from the analogous argument in the opposite limit $T_a \gg T_b$. In fact, it is clear this same mechanism, hence properties (ii) and (iii), apply to *any* quantum “working substance” with an “idle” ground level, and for which $W_{b \rightarrow a}$ always has the same sign.

For property (iv) note that, for high T_b , the populations p_j^b all become close. Due to the symmetry of levels $\pm 2h$, exponentially little work is then extracted in the expansion stroke ($W_{b \rightarrow a} \rightarrow 0$). Thus, $W_{\text{cycle}} < 0$ for sufficiently high T_b .

To understand property (i), note first that when $J = 0$ the spins remain in thermal states throughout the cycle. They must therefore lose entropy when ceding heat, and vice versa. In particular, during a “heater” cycle they would lose entropy in both isochoric strokes. But since there is no other source of entropy, they would then be unable to return to their initial state. Thus, a heater cycle cannot happen. When $J \neq 0$, however, the spins leave equilibrium during the adiabatic strokes. This makes it possible, under some circumstances, for them to *gain* entropy while ceding heat to a thermal bath. (If this idea sounds strange, then consider a system initially in a high-energy eigenstate thermalizing in contact with a low-temperature bath). This is what allows a heater cycle.

Returning now to Eq. (7), note that, for $T_a > T_b$, $W_{b \rightarrow a}$ decreases with J at a faster rate than $|W_{a \rightarrow b}|$. Consider a point just inside the E zone of Fig. 3(a), where $W_{b \rightarrow a} \gtrsim |W_{a \rightarrow b}|$. As J grows, eventually this relation must invert, and W_{cycle} becomes < 0 . By continuity, $Q_a > 0$, $Q_b < 0$ on both sides of the $W_{\text{cycle}} = 0$ border, so this point must in fact become an accelerator [thin green region in Fig. 3(b)]. This process is illustrated in Fig. 4(a). Other transitions, such as from refrigerator to heater to accelerator, can be similarly understood.

In the Appendix we analyze Fig. 3 in greater analytical detail. For example, in Appendix A we prove that counter-rotating engines cannot exist while $J \leq \frac{h_b}{4}$, as in Figs. 3(a) and 3(b), but do whenever $J > \frac{h_a}{4}$, together with the gap mentioned in property (v). In fact, this gap is “direct” (occurs at a specific value of T_b). In Appendix B we derive asymptotic expressions for the interzone boundaries, which help us understand the shapes and disposition of the various zones, in particular property (vi). These expressions also help us deduce asymptotic properties of the engine’s efficiency (see Fig. 5).

VI. EFFICIENCY

Let us finally discuss engine efficiency. Since $n = 1 \notin \mathcal{W}$ then, by Eq. (4), the cycle efficiency when $T_a > T_b$ is

$$\frac{\eta}{\eta_0} = 1 - \frac{q_1^a}{Q_a} = 1 + \frac{8J}{Q_a} \Delta p_1. \quad (9)$$

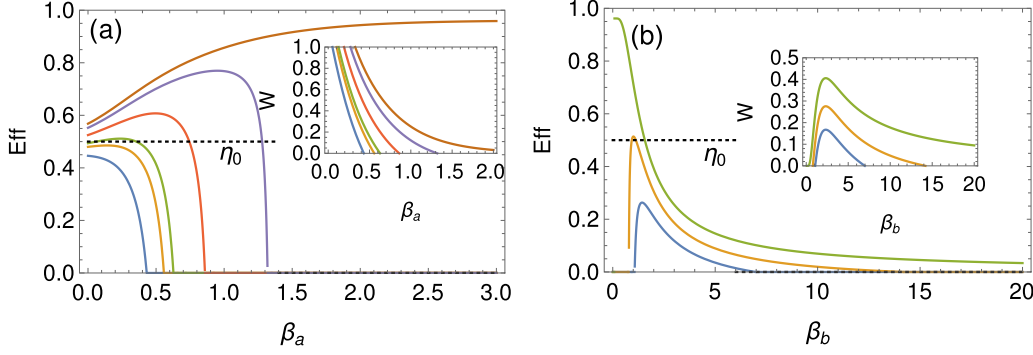


FIG. 5. Efficiency η for fixed J , $h_b = 1$, $h_a = 2$, and various temperature combinations. (a) η vs. $\beta_a \equiv T_a^{-1}$ for $T_b = 0.2, 0.1, 0.08, 0.05, 0.03, 0.001$ (from lower to upper curves) and $J = 0.24$. For $\beta_a \rightarrow 0$, η surpasses the uncoupled efficiency $\eta_0 = 0.5$ if T_b is under the threshold value $\frac{2h_b(h_b-4J)}{\ln 3} = 0.073$ (see Appendix B). It increases even further with β_a (i.e., as the temperature difference between the baths decreases). However, the work output W_{cycle} becomes exponentially small (inset). (b) η vs. $\beta_b \equiv T_b^{-1}$ for $T_a = 0.04, 0.03, 0.003$ (from lower to upper curves) and $J = 0.51$. In this case $\eta > \eta_0$ is achievable with finite work output (inset).

This expression was found, albeit without a physical interpretation, in Ref. [39]. As we have noted, it means that an increase in the efficiency ($\eta > \eta_0$) can happen only when the heat flux $q_1^a = -8J\Delta p_1$ through level 1 is negative, i.e., from the cold to the hot bath (and the increase is in fact proportional to the size of this flux). The precise conditions for this to happen depend however in a complicated way on the values of J and the bath temperatures. We will not explore these details here, but we have found numerically that, for very small values of J , most regimes of operation as an engine [e.g., most of the E zone (red or darker gray region) in Fig. 3(b)] has $\eta > \eta_0$. However, this changes rapidly with J : In Fig. 3(c), only a tiny region close to the origin still has enhanced efficiency, while the entire upper E zone appears to have $\eta < \eta_0$.

We can prove analytically, however, another counterintuitive phenomenon involving the efficiency, illustrated in Fig. 5. In Fig. 5(a) we plot η as a function of $\beta_a \equiv T_a^{-1}$, for various fixed values of T_b and a fixed $J \lesssim \frac{h_b}{4}$. Note first that, as $\beta_a \rightarrow 0$, η tends to a value above η_0 (dashed line) only when T_b is under a certain threshold. Indeed, in Appendix B we prove that the threshold value is $T_b^0 \equiv \frac{2h_b(h_b-4J)}{\ln 3}$. Moreover, we also prove that in these cases the efficiency *increases further* with β_a (i.e., as T_a decreases), as can be seen in the figure [42]. This behavior is contrary to the customary wisdom that a heat engine should become *less* efficient as one decreases the temperature difference across which it operates. Physically what happens is that, as T_a falls, both the overall absorbed heat Q_a and the counterpropagating heat flux through level 1 reduce in magnitude, however the latter does so at a slower rate. By Eq. (9), this increases the efficiency. However, in this regime the extracted work W_{cycle} also falls to 0 exponentially with β_a (see inset).

Another regime exists, however, where a similar effect happens but with W_{cycle} still remaining finite. Consider the counter-rotating engine regime when $T_a \ll T_b$. In this case, as mentioned above in property (iv), for any fixed T_a there is a finite T_b above which no engine is possible. It follows that, as $\beta_b \equiv T_b^{-1}$ is increased from zero, η will at a certain point change from zero to positive and continue to increase, at least for a while. This is verified in Fig. 5(b). As can be seen in

the inset, in these cases W_{cycle} can indeed remain finite for $\eta > \eta_0$.

VII. DISCUSSION

We have presented an alternative mechanism to achieve greater efficiency in quantum thermal machines operating via the Otto cycle. It applies when at least one of the system's energy levels does not couple to the external work source. Moreover, this mechanism also leads to counterintuitive phenomena such as the operation of an engine in both directions of the same cycle, an efficiency increase when the bath temperature difference decreases and situations where the engine can cease to be possible if the hot bath becomes too hot, or the cold bath too cold, or when one of the baths' temperatures lies within a specific range. We have also given an interpretation in terms of separate heat transfers from one bath to another via each system level, showing that some of these effects occur when at least one of these fluxes is in a direction contrary to that of the overall heat flow. Finally, we have illustrated these phenomena in a simple physical model of two spins-1/2 coupled via the Heisenberg interaction.

We emphasize that, unlike in Refs. [4–6,8,9], neither energy-basis coherence nor ergotropy has been exploited here—the system remains always in a passive [7], energy-diagonal state. Moreover, all results for our spin model would still hold if its eigenvectors were all product states—in which case the spins would remain unentangled throughout the cycle. Indeed, they also hold for a three-level (qutrit) model with energies $-8J, \pm 2h$ [43]. This proves the presence of entanglement is not relevant for efficiency gains, confirming indications by previous studies [15–23]. The only quantum aspect of our engine that seems relevant to the phenomena we have described is its discrete energy spectrum [44,45]. Combined with the unitary isoentropic strokes, this allows the working substance to leave thermal equilibrium, which appears to be a key requisite. Note the same could occur in a fully classical stochastic model [46,47] with an effective discrete spectrum and probability-preserving strokes. However,

we could not locate any description in the literature of such phenomena occurring in a classical engine.

Although we have restricted our analysis to specific, idealized scenarios, we expect the phenomenology will generalize. In Appendix C we briefly discuss three possible generalisations, none of which feature “idle” levels. Preliminary findings [48] also indicate most results extend to cycles where nonadiabatic time-dependent strokes do couple different energy eigenstates, resulting in quantum friction [30].

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APPENDIX A: ANALYSIS OF THE COUPLED SPIN OTTO CYCLE

1. Conditions for engines with $T_b > T_a$

As we have noted, one of the remarkable features of this coupled spin system is that it is able to operate as a heat engine in either sense (clockwise or counterclockwise) of the Otto cycle, as depicted in an S - h diagram. However, while a “normal,” counterclockwise engine cycle (running with $T_a > T_b$) can exist for any value $J \geq 0$ of the coupling constant, a clockwise cycle (running with $T_b > T_a$) can only operate as an engine if the coupling is sufficiently strong. More specifically:

Result 1: Suppose $T_b > T_a$. (i) A cycle producing positive work $W_{\text{cycle}} > 0$ is impossible if $0 \leq J \leq h_b/4$. (ii) Conversely, if $J > h_a/4$, then a cycle with $W_{\text{cycle}} > 0$ is possible for all T_b and sufficiently small T_a .

This result is illustrated in the fact that there is no red (darker gray) (E) zone in the bottom right half of Figs. 3(a) and 3(b), but one does appear in Figs. 3(c) and 3(d). Moreover, this bottom E zone runs along the entire T_b axis.

Proof of Result 1: As shown in Eq. (8), the amount of work exchanged in each adiabatic stroke is proportional to the “work function”

$$f_j(h, J, T) = \frac{2 \sinh(2h_j/T_j)}{1 + 2 \cosh(2h_j/T_j) + e^{8J/T_j}}, \quad (\text{A1})$$

where $j \in \{a, b\}$ denotes the starting point of the stroke. It is useful to rewrite this in the form

$$f_j(h, J, T) = g\left(\frac{2h_j}{T_j}, \frac{4J}{h_j}\right), \quad (\text{A2})$$

where

$$g(x, y) = \frac{e^x - e^{-x}}{1 + e^x + e^{-x} + e^{xy}}. \quad (\text{A3})$$

This function, shown in Fig. 6, has the following easily checked properties for $x, y \geq 0$:

$$g(x, y) \geq 0; \quad g(x, y) = 0 \iff x = 0, \quad (\text{A4})$$

$$\lim_{x \rightarrow \infty} g(x, y) = \begin{cases} 0, & y > 1; \\ 1/2, & y = 1; \\ 1, & 0 \leq y < 1. \end{cases} \quad (\text{A5})$$

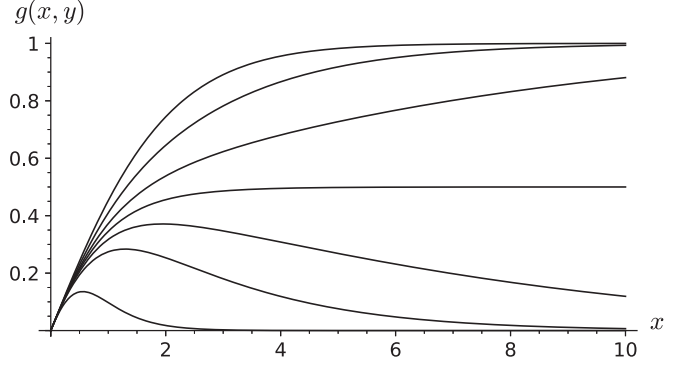


FIG. 6. Work function in the generic form $g(x, y)$ of Eq. (A3), shown as a function of x for various fixed values of y . From top to bottom: $y = 0.1, 0.5, 0.8, 1, 1.2, 1.5, 3$. For $y < 1$, the function increases monotonically toward 1 as $x \rightarrow \infty$. For $y = 1$, it tends asymptotically to $1/2$. For $y > 1$, $g(x, y)$ is single-peaked and tends to 0 as $x \rightarrow \infty$.

$$\frac{\partial g(x, y)}{\partial x} > 0 \text{ for } 0 \leq y \leq 1, \quad (\text{A6})$$

$$\frac{\partial g(x, y)}{\partial x} = 0 \text{ at a single value of } x, \text{ for } y > 1, \quad (\text{A7})$$

$$\frac{\partial g(x, y)}{\partial y} < 0, \quad (\text{A8})$$

$$g(rx, y) > g(x, ry) \text{ for } r > 1. \quad (\text{A9})$$

Suppose then that $T_b > T_a$, and $0 \leq J \leq \frac{h_b}{4}$, i.e., $J = s \frac{h_b}{4}$, for some $0 \leq s \leq 1$. In this case,

$$f_b - f_a = g\left(\frac{2h_b}{T_b}, s\right) - g\left(\frac{2h_a}{T_a}, \frac{s}{r}\right) < 0. \quad (\text{A10})$$

where we used Eq. (A9) and the fact that $r = \frac{h_a}{h_b} > 1$ [49]. By Eq. (7), this means $W_{\text{cycle}} < 0$, so no engine is possible for J in this range.

On the other hand, suppose $J = s \frac{h_a}{4}$, for some $s > 1$. In this case

$$f_b - f_a = g\left(\frac{2h_b}{T_b}, rs\right) - g\left(\frac{2h_a}{T_a}, s\right). \quad (\text{A11})$$

By Eq. (A5), $\lim_{T_a \rightarrow 0} f_a = 0$. Thus, for any fixed T_b we must have $f_b - f_a > 0 \iff W_{\text{cycle}} > 0$ when T_a is sufficiently small ■

2. Temperature gap

Another striking feature of this model, visible in Fig. 3(c), is the appearance of a *temperature gap* for T_a , that is, a range of values $T_a \in (T_{a1}, T_{a2})$ for which the cycle cannot operate at all as an engine—whatever the value of T_b . For T_a within this range, the work extracted during the expansion stroke of the cycle is always less than the one invested in the compression stroke.

We will now prove that such a gap always exists if the coupling J is sufficiently high. In fact, borrowing the terminology of solid-state physics, it is always a “direct gap,” in the sense that these two extremes of the gap range occur for the same value of T_b . Graphically, the lowest point of the “upper E

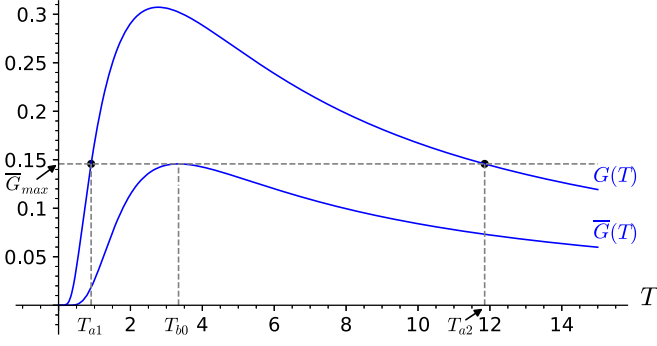


FIG. 7. Functions $G(T)$ and $\bar{G}(T)$ defined in Eq. (A13). The highlighted dots show the two points where $G(T)$ reaches the maximum value $\bar{G}_{\max}$ of $\bar{G}(T)$. The interval $T_{a1} < T < T_{a2}$ where $G(T) > \bar{G}_{\max}$ corresponds to the gap in T_a where no engine can exist in Fig. 3(c). The temperature T_{b0} which maximizes $\bar{G}(T)$ is the point in T_b where the gap occurs. The parameters $h_a = 2$, $h_b = 1$, $J = 0.7$ are the same here as in that figure. For these values, $T_{a1} = 0.91$, $T_{b0} = 3.34$, and $T_{a2} = 11.85$.

zone” is always directly above the highest point of the “lower E zone.”

Result 2: For $J > h_a/4$, there exists a range (T_{a1}, T_{a2}) of values of T_a for which $W_{\text{cycle}} < 0$, for any value of T_b . Furthermore, given h_a, h_b, J , there exists a temperature T_{b0} such that

- (i) T_{b0} is the value of T_b that maximizes the work $W_{b \rightarrow a}$ extracted during the expansion stroke,
- (ii) $T_{b0} \in (T_{a1}, T_{a2})$, and
- (iii) $W_{\text{cycle}}(T_a, T_b) = 0$ for $(T_a, T_b) = (T_{a2}, T_{b0})$ and $(T_a, T_b) = (T_{a1}, T_{b0})$. In other words, the gap in T_a is “direct” and occurs at $T_b = T_{b0}$.

Proof of Result 2: We will base our analysis purely on the general mathematical properties (A4)–(A9) of $g(x, y)$. Note first that the borders of the E zones are defined by the condition $W_{\text{cycle}} = 0$. For the case $J > h_a/4$, Eq. (A11) shows this condition is equivalent to

$$g\left(\frac{2h_a}{T_a}, s\right) = g\left(\frac{2h_b}{T_b}, rs\right), \quad (\text{A12})$$

where $r, s > 1$. Since we are assuming fixed h_a, h_b, J here, it is helpful to consider each side of this equation as a function only of T , e.g., to redefine

$$g\left(\frac{2h_a}{T_a}, s\right) \equiv G(T_a); \quad g\left(\frac{2h_b}{T_b}, rs\right) \equiv \bar{G}(T_b). \quad (\text{A13})$$

Let us now analyze the temperature ranges for which Eq. (A12) may or may not have solutions. Since $r, s > 1$, Eqs. (A4), (A5), (A7) imply that $G(T)$ and $\bar{G}(T)$ are single-peaked in T , and tend to 0 for $T \rightarrow 0, \infty$ (see Fig. 7). Let T_{b0} be the temperature where $\bar{G}(T_{b0})$ achieves its peak $\bar{G}_{\max} \equiv \max_T \bar{G}(T)$. Comparing the definitions of $\bar{G}$ in Eq. (A13) and $W_{b \rightarrow a}$ in Eq. (7), using also Eq. (A2), we see this is the same as property (i).

Now Eq. (A8) implies that

$$\bar{G}_{\max} = g\left(\frac{2h_b}{T_{b0}}, rs\right) < g\left(\frac{2h_b}{T_{b0}}, s\right) = G(rT_{b0}) \leq G_{\max}, \quad (\text{A14})$$

where $G_{\max} \equiv \max_T G(T)$. Since $G(T)$ is single-peaked, it follows that, given any $T_b > 0$, Eq. (A12) will always be satisfied for exactly two values of T_a , that we will call $T_{a+}(T_b) > T_{a-}(T_b)$. (These solutions describe the upper and lower E zone borderlines in Fig. 3(c). In particular, if we define

$$T_{a1} \equiv T_{a-}(T_{b0}); \quad T_{a2} \equiv T_{a+}(T_{b0}),$$

then $G(T_{a1,2}) = \bar{G}_{\max} < G_{\max}$. Since $G(T)$ is single-peaked, $G(T_a) > \bar{G}_{\max} \iff T_a \in (T_{a1}, T_{a2})$. But this implies there is no solution for Eq. (A12) for T_a in this interval. More specifically, within this gap

$$W_{\text{cycle}} \propto [\bar{G}(T_b) - G(T_a)] < [\bar{G}_{\max} - G(T_a)] < 0, \quad \forall T_b.$$

Conversely, for any $T_a > T_{a2}$ or $< T_{a1}$, an analogous argument shows there will be two values of T_b that solve Eq. (A12). However, since $\bar{G}$ is single-peaked, for $T_a = T_{a2}$ or $T_a = T_{a1}$ the only solution is $T_b = T_{b0}$. Thus we obtain property (iii).

We still need to prove property (ii). For the upper bound: by Eq. (A14), $G(rT_{b0}) > G(T_{a2}) = G(T_{a1})$. Since G is single peaked, and by definition $T_{a2} > T_{a1}$, this implies $T_{a2} > rT_{b0} > T_{b0}$.

For the lower bound: Equation (A9) implies that, in fact,

$$G(T) > \bar{G}(T), \quad \forall T > 0.$$

as can be seen in Fig. 7. But then $G(T_{b0}) > \bar{G}(T_{b0}) = G(T_{a1}) = G(T_{a2})$. Since G is single-peaked, this means that $T_{a1} < T_{b0} < T_{a2}$. Note that we reobtain also the upper bound in this way. However, as seen above, the latter is independent of property (A9). ■

In conclusion, the temperature gap in T_a appears due to the fact that, for large-enough coupling J , the work function $f(h, J, T)$ becomes single-peaked in h/T . Physically, this occurs because, for large J , the ground state of this model becomes the level $-8J$, which does not contribute to the work exchanges in the adiabatic strokes, but which contains more and more of the system’s population as $T \rightarrow 0$. The appearance of this gap can therefore be argued to be a quantum effect, since such a population accumulation in a single discrete state does not happen in classical systems.

It is worth emphasizing that, in the demonstration of Result 2, we only made use of properties (A4)–(A9) of the work function, and not of the specific form of $f(h, J, T)$. Since these are quite generic, we can expect that other, more complex models with similar characteristics will also exhibit a temperature gap where no engine operation is possible. An interesting open problem is to investigate more carefully to what extent properties such as Eqs. (A4)–(A9) are mandated by general constraints such as the second law, irrespective of the details of the Hamiltonian. For example: note that, in the demonstration above, the only place where property (A9) was required was to prove $T_{a1} < T_{b0}$. In the absence of this property, a direct gap would still exist, but it might be entirely within the region where $T_a > T_b$. This would, however, imply an engine existing for $T_a = T_b$, which is forbidden by the second law, for we would be extracting positive work from a single thermal reservoir. We therefore conclude that if a

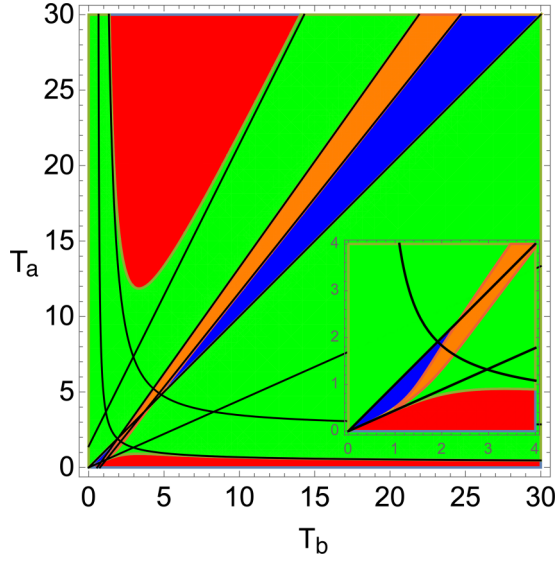


FIG. 8. Asymptotic approximations for the borders of the various operation zones of the system. The system parameters are $h_a = 2$, $h_b = 1$, $J = 0.7$ [same as in Fig. 3(c)]. As before, the refrigerator (R) zones are in blue (darkest gray), the engine (E) zones in red (darker gray), the heater (H) in orange (lighter gray) and the accelerator (A) in green (lightest gray). The black lines are our asymptotic approximations to the various interzone boundaries. For the upper E zone, these are given by Eq. (B7) (left side) and Eq. (B5) (right side). For the lower E zone, they are given by Eq. (B9) (left side) and Eq. (B11) (right side). For the A-H, H-R, and R-A borders, they are given respectively by Eqs. (B15), (B16), and (B21). The inset shows a closer view of the bottom left corner, omitting Eqs. (B15) and (B16) for clarity. Note in particular the second R zone, corresponding to a “counter-rotating” refrigerator ($T_b > T_a$).

system satisfies properties (A4)–(A8), the second law requires it to also satisfy property (A9).

APPENDIX B: ASYMPTOTIC ANALYSIS

In this section we study how the system behaves when the temperatures T_a , T_b , become relatively “large” or “small” in comparison to each other or to other parameters. We use the resulting asymptotic formulas to understand the boundaries between the various zones appearing in the diagrams of Fig. 3 (see Fig. 8 for greater detail). Another application is to study the asymptotic behavior of the heat engine efficiency η in the limit of large T_a . In particular, we demonstrate that, in this limit, η can increase as T_a decreases, for fixed T_b .

Our analysis is based on the following simple approximations:

(i) For “large” temperatures, satisfying $2h_j/T_j \sim 8J/T_j \sim \varepsilon \ll 1$, we can express the work function in Eq. (A1), and more generally the probabilities p_1^j appearing in the text, as power series in ε :

$$f_j(h, J, T) = \frac{h_j}{T_j} - \frac{2h_j J}{T_j^2} + O(\varepsilon^3), \quad (\text{B1})$$

$$p_1^j = \frac{1}{4} \left[1 + \frac{6J}{T_j} + \frac{12J^2 - h_j^2}{T_j^2} \right] + O(\varepsilon^3). \quad (\text{B2})$$

(ii) For “small” temperatures $T_j \ll 8J, 2h_j$, the appropriate approximations are

$$f_j(h, J, T) \simeq \frac{1}{1 + \exp\left(\frac{8J - 2h_j}{T_j}\right)}, \quad (\text{B3})$$

$$p_1^j(h, J, T) \simeq \frac{1}{1 + \exp\left(\frac{2h_j - 8J}{T_j}\right)}. \quad (\text{B4})$$

In other words, in this limit these functions behave as (complementary) step functions in J , with a sharp cutoff for $J > h_j/4$.

1. Asymptotic zone boundaries

Our first application is to obtain asymptotic expressions for the boundaries between the various operation regimes of the system. Figure 8 above illustrates the result of these calculations, which are detailed below. The figure is an expanded version of Fig. 3(c), with added lines corresponding to our asymptotes. We can see that agreement with the exact zone boundaries is generally extremely good, even for temperatures well within the depicted range.

a. Boundary of the upper E zone

The boundaries of the E zones are defined by the condition $W_{\text{cycle}} = 0$, which is equivalent to $f_a = f_b$ by Eq. (7). For the “upper” ($T_a > T_b$) E zone, we can solve this equation for T_a in the limits where (i) T_a and T_b are both “large” or (ii) T_a is “large” and T_b is “small.”

(i) Suppose $T_a \gtrsim T_b \gg 2J, 2h_a, 2h_b$. In this case it is possible to find T_a by an iterative (perturbative) method. Using Eq. (B1), we solve the equation $f_a = f_b$ to $O(\varepsilon)$, then substitute the solution into the equation to $O(\varepsilon^2)$, and so forth.

In this manner, it is possible in principle to obtain a complete series expression for the boundary line, valid in the limit of large T_b . The solution correct up to $O(\varepsilon)$, which is sufficient for our purposes, is

$$T_a = rT_b + 2J(r - 1). \quad (\text{B5})$$

Note that when $J = 0$ (no coupling), we recover asymptotically the exact E-zone boundary [Fig. 3(a)]. For $J > 0$, Eq. (B5) shows that the boundary remains asymptotically a straight line, with the same inclination as previously, but shifted upwards by an amount proportional to J . Physically this means that, for a given “large” temperature T_b , the temperature T_a required to run an engine increases with the coupling strength. This can indeed be seen in Figs. 3(b) and 3(c). In Fig. 3(d) the upper E zone exists but is not visible—it has shifted up completely out of the depicted area, since J is so large.

(ii) For $T_b \ll 8J, 2h_b, 2h_a \ll T_a$, the boundary condition, correct to $O(\varepsilon) \sim h_a/T_a \sim J/T_a$, becomes

$$\frac{h_a}{T_a} \left(1 - \frac{2J}{T_a} \right) = \left[1 + \exp\left(\frac{8J - 2h_b}{T_b}\right) \right]^{-1}. \quad (\text{B6})$$

Inverting both sides and using the fact that, to same order in ε , $\frac{1}{(1-\frac{2J}{T_a})} = 1 + \frac{2J}{T_a}$, we obtain

$$T_a = -2J + h_a \left[1 + \exp\left(\frac{8J - 2h_b}{T_b}\right) \right]. \quad (\text{B7})$$

When $J > h_a/4$ ($> h_b/4$), Eq. (B7) implies that the minimum temperature T_a necessary for an engine to be possible *increases exponentially* as $T_b \rightarrow 0$. This remarkable property, which can be clearly seen in Fig. 8, can be physically understood as follows: Recall that $J > h_a/4$ implies that the ground state is the level $-8J$, which does not take part in work exchanges. After the system thermalizes with the very cold bath at $T = T_b$, the population of all other levels will be exponentially small, so the same will be true for the amount of work extracted during the dilation stroke. The only way to have an engine in these conditions is to expend an even smaller amount of work in the compression stroke. This is only possible if the populations of levels $\pm 2h$ are exponentially close to each other before the stroke, which in turn requires an exponentially large temperature T_a .

On the other hand, if $J < h_b/4$, then the exponential in Eq. (B7) goes quickly to zero for small T_b , and the right-hand side of this expression becomes $< h_a$. However, this contradicts our starting hypothesis that $T_a \gg h_a$, indicating that there is in fact no physical solution. Indeed, as already noted in the main text [and seen in Figs. 3(a) and 3(b)], in this case the *E* zone has no boundary at all for small T_b (an engine always becomes possible as $T_b \rightarrow 0$, for any T_a).

b. Boundary of the lower *E* zone

For the “lower” *E* zone ($T_b > T_a$), we again look for solutions to $f_a = f_b$, but in the limits where (i) T_a is “small” and T_b is “large” or (ii) T_a and T_b are both “small.”

(i) For $T_a \ll 8J$, $2h_b$, $2h_a \ll T_b$ we obtain an equation analogous to Eq. (B6) but with a, b exchanged:

$$\frac{h_b}{T_b} \left(1 - \frac{2J}{T_b} \right) = \left[1 + \exp\left(\frac{8J - 2h_a}{T_a}\right) \right]^{-1}. \quad (\text{B8})$$

Solving for T_a this time, using the same approximation, we obtain

$$T_a = \frac{8J - 2h_a}{\ln(T_b + 2J - h_b) - \ln(h_b)}. \quad (\text{B9})$$

Consistently with Result 1, we see that a solution for an engine in this regime requires $J > h_a/4$.

Note that Eq. (B9) is a (slowly) *decreasing* function of T_b . As discussed in the main text, this implies the very counterintuitive property that, for any fixed T_a , there is a finite value of $T_b > T_a$ above which the cycle ceases to function as an engine.

(ii) For $T_a, T_b \ll 2h_b$, $2h_a < 8J$, we use Eq. (B3) for both work functions, obtaining

$$\frac{8J - 2h_a}{T_a} = \frac{8J - 2h_b}{T_b} \quad (\text{B10})$$

or

$$T_a = r_E T_b \quad (\text{B11})$$

TABLE I. The four operational regimes allowed by the first and second laws for a thermal machine operating between two thermal reservoirs [14]. Note we use opposite sign conventions for heat and work.

	W_{cycle}	Q_{hot}	Q_{cold}
Engine	>0	>0	<0
Accelerator	<0	>0	<0
Refrigerator	<0	<0	>0
Heater	<0	<0	<0

with

$$r_E = \frac{4J - h_a}{4J - h_b}. \quad (\text{B12})$$

Note $0 < r_E < 1$ for $J > h_a$.

c. Boundaries of the *H*, *R*, and *A* zones

It is worth recalling the conditions defining each zone, which are summarized in Table I above. In terms of energy exchanges, the difference between a *heater* and an *accelerator* is in the sign of Q_{hot} . The boundary between the *H* and *A* zones is therefore defined by the condition $Q_{\text{hot}} = 0$. Similarly, the difference between a heater and a refrigerator is in the sign of Q_{cold} , so the boundary between the *H* and *R* zones is defined by the condition $Q_{\text{cold}} = 0$. For $T_a > T_b$, $Q_{\text{hot}} = Q_a$ and $Q_{\text{cold}} = Q_b$. From Eq. (7), these borders are thus described respectively by the equations

$$8J(p_1^b - p_1^a) = 2h_a(f_a - f_b), \quad (\text{B13})$$

$$8J(p_1^b - p_1^a) = 2h_b(f_a - f_b). \quad (\text{B14})$$

For $2h_j/T_j \sim 8J/T_j \sim \varepsilon \ll 1$, we can use again an iterative method, as with Eq. (B5) above. The solutions to these equations have the linear forms

$$T_a^{AH} = r_{AH} T_b + s_{AH} + O(\varepsilon), \quad (\text{B15})$$

$$T_a^{HR} = r_{HR} T_b + s_{HR} + O(\varepsilon). \quad (\text{B16})$$

where the angular coefficients are, respectively,

$$r_{AH} = \frac{6J^2 + h_a^2}{6J^2 + h_a h_b}, \quad (\text{B17})$$

$$r_{HR} = \frac{6J^2 + h_a h_b}{6J^2 + h_b^2}, \quad (\text{B18})$$

while the linear coefficients are, respectively,

$$s_{AH} = -J \frac{12J^2(r_{AH}^2 - 1) - r_{HR}^2(h_a(2h_a + h_b) + 3h_a^2)}{6J^2 + h_a^2}, \quad (\text{B19})$$

$$s_{HR} = -J \frac{12J^2(r_{HR}^2 - 1) + h_a(2h_b + h_a) - 3h_b^2 r_{HR}^2}{6J^2 + h_a h_b}. \quad (\text{B20})$$

Note that, for $T_b > T_a$, the roles of Q_a, Q_b invert, so the solutions for the two boundaries are merely swapped (T_a^{AH} would now have the expression in Eq. (B16), and vice versa). However, since r_{AH} and r_{HR} are both >1 , these solutions are incompatible with the initial hypothesis ($T_b > T_a$). This

implies that, for large $T_a < T_b$, there can be no refrigerator or heater.

Finally, the difference between an *accelerator* and a *refrigerator* is of a different nature. In both cases one bath loses heat while the other one gains; the difference is whether the bath losing heat is the hotter or the colder one. Thus, the boundary between these zones is not marked by a change of sign in either bath's heat exchange, but in the difference of their temperatures. In other words, in this case the boundary is the line

$$T_a^{AR} = T_b, \quad (\text{B21})$$

irrespective of any asymptotic limits. Note this holds both for refrigerators rotating in the “ordinary” sense (clockwise, $T_a > T_b$), and in the “counter-rotating” sense (counterclockwise, $T_b > T_a$).

It is easy to verify that

$$r > r_{AH} > r_{HR} > 1. \quad (\text{B22})$$

This shows that, for large-enough T_b , if we start from $T_a = T_b$ and increase T_a , then we must obtain in succession a refrigerator, a heater, an accelerator and finally an engine—which is indeed what we observe in Fig. 8. In addition, it also shows that, for large-enough T_b , a heater or refrigerator are only possible if $T_a > T_b$ (equivalently, for large-enough $T_b > T_a$ only an accelerator or an engine are possible).

d. Strong-coupling limit

If J is large compared to h_a, h_b , then it is clear from Eqs. (B12), (B17), and (B18) that r_E, r_{AH} , and r_{HR} all tend to 1. As a consequence (i) both the H and R zones will collapse into an extremely narrow range around the line $T_a = T_b$ [property (vi) in Sec. IV]. In addition: (ii) The lower E -zone border will also accompany this line for low T_b , and as a consequence this zone will increase in size. These effects can be seen in Fig. 3(d).

2. Asymptotic efficiency

Another application of Eqs. (B1)–(B4) is to study the efficiency η in the asymptotic limits where the bath temperatures T_a and/or T_b become very large or very small.

As we have shown in Fig. 5(a), as $T_a \rightarrow \infty$ (i.e., as $\beta_a \rightarrow 0$), η tends to a value above η_0 (dashed line) only when T_b is under a certain threshold. To see why, suppose

$$T_b \ll 8J < 2h_b < 2h_a \ll T_a. \quad (\text{B23})$$

In this case, we obtain, up to an error of order ε ,

$$W_{\text{cycle}} = \frac{2(h_a - h_b) \exp\left[\frac{2h_b - 8J}{T_b}\right]}{1 + \exp\left[\frac{2h_b - 8J}{T_b}\right]} \quad (\text{B24})$$

$$Q_a = \frac{8J + 2h_a \exp\left[\frac{2h_b - 8J}{T_b}\right]}{1 + \exp\left[\frac{2h_b - 8J}{T_b}\right]} - 2J, \quad (\text{B25})$$

so

$$\eta = \frac{2(h_a - h_b) \exp\left[\frac{2h_b - 8J}{T_b}\right]}{6J + 2(h_a - J) \exp\left[\frac{2h_b - 8J}{T_b}\right]}. \quad (\text{B26})$$

Choose a fixed value $J = \frac{h_b}{4}(1 - \delta)$ for some $\delta < 1$. Substituting above, we obtain

$$\begin{aligned} \eta &= \frac{2(h_a - h_b)e^{\frac{2h_b\delta}{T_b}}}{\frac{3}{2}h_b(1 - \delta) + \left[2h_a - \frac{1}{2}h_b(1 - \delta)\right]e^{\frac{2h_b\delta}{T_b}}} \\ &= \frac{\eta_0 e^{\frac{2h_b\delta}{T_b}}}{\frac{3}{4r}(1 - \delta) + \left[1 - \frac{1}{4r}(1 - \delta)\right]e^{\frac{2h_b\delta}{T_b}}}. \end{aligned} \quad (\text{B27})$$

It follows that $\eta > \eta_0$ if and only if

$$\frac{3}{4r}(1 - \delta) < \frac{1}{4r}(1 - \delta)e^{\frac{2h_b\delta}{T_b}} \quad (\text{B28})$$

or

$$T_b < \frac{2h_b\delta}{\ln 3} \quad (\text{B29})$$

(for consistency with the hypothesis that $T_b \ll 2h_b$, we see that we must in fact choose $\delta \ll 1$).

In other words, given a sufficiently high coupling strength, there exists a finite threshold for T_b , under which the engine efficiency converges to a higher than classical value in the limit of high T_a . This can be seen in Fig. 5(a) above. For the parameters in that figure ($J = 0.24, h_a = 2, h_b = 1$), it is easy to check that $\delta = 0.04$ and the threshold value is $T_b \sim 0.073$, which matches well with the exact curves. It is also worth noting that, in this limit, W_{cycle} is not negligible; indeed $W_{\text{cycle}} \rightarrow 2(h_a - h_b)$ for $T_b \ll 2h_b\delta$.

Moreover, it can also be noted from the figure that, whenever T_b is below the threshold, the efficiency actually *increases* further with β_a (ie, as T_a decreases). Let us now demonstrate that this does indeed happen as T_a is reduced from infinity. For this, is useful to look at $\lim_{T_a \rightarrow \infty} \frac{\partial \eta}{\partial \beta_a}$, where $\beta_a \equiv 1/T_a$ is the inverse temperature. We wish to show that this limit is strictly positive for T_b satisfying Eq. (B29). (Directly calculating $\lim_{T_a \rightarrow \infty} \frac{\partial \eta}{\partial T_a}$ does not give the same information, since in this limit $\frac{\partial \eta}{\partial T_a} = -\frac{1}{T_a^2} \frac{\partial \eta}{\partial \beta_a} \rightarrow 0$).

Going back then to the exact equations for Q_a and W_{cycle} [Eq. (7)], replacing $J = \frac{h_b}{4}(1 - \delta)$, and taking the appropriate limits, it is possible to show after some algebra that

$$\lim_{T_a \rightarrow \infty} \frac{\partial \eta}{\partial \beta_a} = \frac{\eta}{W_{\text{cycle}}} \left\{ \eta \left[2h_a^2 + \frac{3}{4}h_b^2(1 - \delta)^2 \right] - 2h_a(h_a - h_b) \right\}. \quad (\text{B30})$$

Using the asymptotic expression for η in Eq. (B27), we find that the term in square brackets is positive if

$$\exp\left[\frac{2h_b\delta}{T_b}\right] > \frac{3}{1 + \frac{3}{2r}(1 - \delta)}. \quad (\text{B31})$$

In other words:

$$\lim_{T_a \rightarrow \infty} \frac{\partial \eta}{\partial \beta_a} > 0 \iff T_b < \frac{2h_b\delta}{\ln 3 - \ln\left[1 + \frac{3}{2r}(1 - \delta)\right]}. \quad (\text{B32})$$

Since this upper bound is strictly larger than the one in Eq. (B29), we arrive at our desired result.

For the parameters in Fig. 5(a), the threshold in Eq. (B32) equals approximately 0.144. It can be seen that η does indeed

initially increase with β_a for all values of T_b below this value [including those under the smaller threshold in Eq. (B29)].

Of course, for sufficiently low T_a an engine becomes impossible, so the efficiency must go to zero (at best, this must happen for $T_a < rT_b$). This implies that the efficiency must reach a maximum at some finite T_a , which is what we observe in the figure.

In summary, we have proven

Result 3: Given a sufficiently high coupling strength $J = \frac{\hbar b}{4}(1 - \delta)$, with $\delta \ll 1$, and a sufficiently low cold bath temperature $T_b \ll 8J$, and taking the limit $T_a \rightarrow \infty$, then the engine efficiency η converges to a value higher than η_0 if and only if T_b is under the threshold given by Eq. (B29). In addition, in these circumstances η increases further as T_a is reduced from infinity, reaching a maximum at some finite T_a .

APPENDIX C: EXTENSIONS OF THE “IDLE LEVEL” SCENARIO

In the Otto engine scenario we discuss, levels can be divided into two groups: They are either “idle,” independent of the adiabatic parameter λ , or else vary with λ according to Eq. (3). This situation, which we will now refer to as the “idle scenario,” is intended as a minimal departure from the uniform scenario studied in Refs. [26,27,36,37], where all levels obey this second property (i.e., all gaps scale proportionally to λ). In those cases, the efficiency always equals the standard value $\eta_0 = 1 - \lambda_b/\lambda_a$. As we have shown, the presence of even a single “idle” level allows a host of interesting effects to appear, including efficiency beyond η_0 , counter-rotating engines, a temperature gap where no engine is possible, etc.

It by no means follows, however, that the idle scenario is the only one where such phenomena can occur, or that the presence of idle levels is crucial for them in any fundamental sense. In this section, we seek to emphasize this point by briefly sketching three extensions of the idle scenario, none of which feature idle levels, but where many of the phenomena we have pointed out will continue to occur. This is not intended as a thorough exploration of these scenarios, let alone an exhaustive list of possibilities, but as an illustration and perhaps a starting point for further studies.

1. Extension by continuity

A first, simple, possibility is to slightly deform the idle scenario. For example, suppose we introduce a small adiabatic shift to each (originally) idle level, so that they cease to be idle. Physically, this means these levels are now very weakly coupled to the work source or sink responsible for the time dependence during the adiabatic strokes. The shifts do not all have to be equal, nor to apply to the “working” levels.

Now, by continuity, for sufficiently small deformations most properties of the original engine must still remain true. For instance, if $\eta > \eta_0$ before the deformation, or if the output work is $W > 0$ in a counter-rotating cycle, then these properties will continue to hold for a slightly deformed engine—even though the latter will no longer have true idle levels, nor the same energy gaps. How large these deformations can be made will of course depend on details of each specific Hamiltonian and adiabatic shifts. An analogous

argument can also be made with respect to scenarios with imperfect thermalization in the isochoric strokes.

2. Extension by symmetry

It seems to be a widely known (but, to our knowledge, unpublished) fact that idealised Otto engines of the kind we discuss, i.e., with full thermalization and perfect adiabatic strokes, possess the following symmetry [50]: The heat exchanges with each bath, and hence the total output work, are invariant under *independent* global energy shifts at either end of the adiabatic strokes. More precisely:

Theorem: Consider an ideal Otto engine E , satisfying Eqs. (1) and (2), with energy levels $\{E_n^a\}$ at $\lambda = \lambda_a$ and $\{E_n^b\}$ at $\lambda = \lambda_b$, respectively. Consider also another Otto engine E' , identical except for the energy shifts $E_n'^a = E_n^a + \delta_a$, $E_n'^b = E_n^b + \delta_b$, $\forall n$, where in general $\delta_a \neq \delta_b$. Then $Q_a = Q'_a$, $Q_b = Q'_b$, and $W_{\text{cycle}} = W'_{\text{cycle}}$.

Proof: The case $\delta_a = \delta_b$ is trivial, since it corresponds to a global energy shift of the entire scenario. For similar reasons, we can always, without loss of generality, choose $\delta_b = 0$, i.e., analyze only the effect of $\delta_a \neq 0$ (we assume this from now on). Note now that $p_n'^j = p_n^j$, $\forall n, j$. The result then follows straightforwardly from Eq. (1), and from the fact that $W'_{\text{cycle}} = Q'_a + Q'_b$. ■

This symmetry implies, in particular, that engines E and E' will both behave exactly the same with respect to efficiency, regimes of operation, sense of rotation of the cycle, and more generally all the phenomena reflected in Fig. 3.

Nevertheless, a global shift in E_n^a is not a complete symmetry of the engine. It does affect the separate work exchanges $W_{a \rightarrow b}$ and $W_{b \rightarrow a}$ in each adiabatic stroke (even though their sum is invariant). For instance

$$\begin{aligned} W'_{a \rightarrow b} &= \sum_n p_n'^a (E_n'^b - E_n'^a) \\ &= \sum_n p_n^a (E_n^b - E_n^a - \delta_a) \\ &= W_{a \rightarrow b} - \delta_a. \end{aligned}$$

This shows the original and “shifted” engines describe *physically different* situations, i.e., they are indeed different engines with similar properties.

Note now finally that if engine E operates in the “idle scenario,” then E' will generally *not*: For nearly all values of δ_a , it will in fact have no idle levels whatsoever.

3. Extension by linearity

Finally, consider an engine whose levels can be divided into two groups (A and B), each of which shifts adiabatically according to Eq. (3), but with different linear coefficients, as follows:

$$\begin{aligned} \text{Group A: } E_{n,A}(\lambda) &= c_{n,A}\lambda \\ \text{Group B: } E_{n,B}(\lambda) &= \delta + c_{n,B}\lambda, \end{aligned}$$

where all $c_{n,j} \neq 0$ and where we can choose the constant $\delta > 0$ without loss of generality. Thus, for small-enough λ , Group B levels are all higher than those in Group A.

There are no idle levels in this scenario either. Nevertheless we can apply much of the same reasoning from that case. For example: Following a derivation analogous to the one leading to Eq. (4), we now obtain that, for $T_a > T_b$ and $Q_a > 0$, an engine will have efficiency

$$\frac{\eta}{\eta_0} = 1 - \delta \frac{\sum_{n,B} \Delta p_{n,B}}{Q_a}, \quad (\text{C1})$$

where $\Delta p_{n,B} = p_{n,B}^a - p_{n,B}^b$. Once again, an increase with respect to the standard Otto efficiency η_0 is in principle possible, with no presence of energy-basis coherence, etc. The condition for this increase to occur in this new scenario is, however, different, namely:

$$\sum_{n,B} \Delta p_{n,B} < 0. \quad (\text{C2})$$

In other words, this engine will have a greater-than-standard efficiency only if the overall population of the B levels decreases as the temperature increases from T_b to T_a , and λ increases from λ_b to λ_a . In particular, suppose, for simplicity, that the engine operates between values of λ that are sufficiently small that no level crossings occur between Group A

and Group B levels. Then we need the population of the higher (B) levels to decrease when the temperature increases.

Achieving this is not entirely obvious—for example, for a fixed Hamiltonian, the highest level of a finite-dimensional quantum system always *gains* population with increasing temperature. However, the reverse is not impossible when the variation in λ is also taken into account. For instance, if the highest levels have large positive $c_{n,B}$ (and thus shift significantly upward as λ increases from λ_b to λ_a), then their populations at $T_a > T_b$ may well become smaller than at T_b , despite the increase in temperature. A simple three-level example where this indeed happens (with, e.g., the highest level alone constituting “Group B,” and the other two “Group A”) is as follows:

$$\begin{aligned} E_{1,A}(\lambda) &= \lambda \\ E_{2,A}(\lambda) &= 2\lambda \\ E_{1,B}(\lambda) &= 1 + 4\lambda. \end{aligned}$$

It can be easily checked that, for example, if $T_b = 1$, $T_a = 4$, $\lambda_b = 1$, $\lambda_a = 2$, then p_{1B} decreases from about 1.3% to about 1.1% as λ and T both increase from b to a .

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